Axis-Squeeze and Multirouting Scale-Adaptive Fusion Network for Remote Sensing Images Object Detection

Yan Chen, Qi Gu, Xiaofeng Wang, Xinlu Shi, Lixiang Xu, and Chen Zhang

***Abstract*—Complicated background and small object issues are the primary challenges currently confronting remote sensing object detection (RSOD). To tackle the aforementioned issues in RSOD, we propose an axis-squeeze and multirouting scale-adaptive fusion network (AMSFNet). In this network, a unidirectional multiscale coupling module (UMCM) is designed to enhance the object feature extraction capabilities and improve the accuracy of small object detection. Furthermore, utilizing the axis-squeeze and detail enhancement approach, we construct a squeeze-enhanced axial attention module that specifically targets background disruption, which can efficiently aggregate global and local information and reduce the impact of intricate background noise on the object. To improve the synergy between features of varying sizes and ensure that the model can allow for a compromise between detecting small and large targets simultaneously, a multirouting SAF approach is proposed, which combines fine-grained features with high-level features using multiple feedforward connections. The proposed approach has been shown to be efficient by ablation and comparison experiments performed on three public benchmark datasets: RSOD, VisDrone, and DIOR. AMSFNet achieves a mean average precision (mAP) of 95.4%, 88.7%, and 36.6% on the RSOD, DIOR, and VisDrone datasets, respectively.**

***Index Terms*—Object detection, remote sensing images (RSIs), scale-adaptive fusion (SAF), small object, squeeze-enhanced axial attention.**

I. Introduction

REMOTE sensing object detection (RSOD) is well recognized in the field of remote sensing image (RSI) analysis, which is commonly utilized in traffic management, safety search, military surveillance, and ecological protection [1], [2], [3]. In contrast to natural scene images, sensing images are captured from a top-down perspective, resulting in considerable scale differences, a high density of small objects, and complex backgrounds [4], as shown in Fig. 1. Conventional spectral-based and object-oriented object detection algorithms often exhibit limited performance due to the challenges in proficiently capturing and integrating multilevel and multiscale features in RSIs, which leads to a reduction in detection precision and recall.

In recent years, deep learning methods represented by convolutional neural networks (CNNs) have been widely applied. The existing dominant deep learning object detection methods can be categorized into one-stage and two-stage. One-stage detection algorithms mainly include YOLO [5], [6], [8], and SSD [9], [10]. By way of illustration, YOLO partitions the image into several grid cells. Within each grid cell, objects are detected and the categories and positions of bounding boxes are consequently predicted. The one-stage design simplifies the processing and enhances the detection speed, although it lacks certain accuracy. In contrast to one-stage detection, two-stage detection methods like Faster R-CNN [11] first employ a region proposal network to create candidate regions, then perform classification and bounding box regression. This approach consists of two separate stages: the initial step is to extract the region of interest, while the subsequent stage is dedicated to classification and localization. It has high accuracy and can handle complex scenarios but slower in detection speed. The aforementioned methods rely on a substantial number of anchor boxes, which in turn involves several hyperparameters and computations. In addition, the dimensions of the anchor box must be clustered based on the dataset, which limits flexibility. Hence, several anchor-free detection methods have also been proposed in recent years, such as CenterNet [13], CornerNet [12], and FCOS [14]. For instance, CenterNet converts the object detection problem into the identification of keypoint detection. It achieves detection by predicting the center points of objects along with their width and height. In contrast, CornerNet adopts a different approach by detecting the two corner points of an object to achieve detection. FCOS discards predefined anchor boxes by predicting the location and category of objects through regression at each pixel in the feature map. This design enhances detection flexibility and eliminates performance limitations associated with anchors. Significantly, YOLOX [15] introduces a decoupled head that efficiently utilizes semantic and positional information to enhance the convergence. Moreover, the

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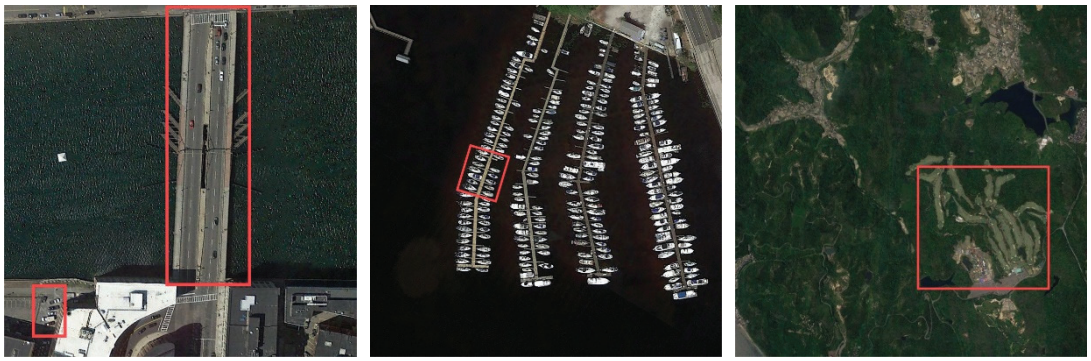


Fig. 1. Samples of object detection in the typical RSI, where (a) shows a significant size variance between the car and bridge; (b) presents the dense small objects; and (c) demonstrates the various and complicated background, where a GC has a significant level of similarity to the surrounding environment, resulting in interference.

trick integrated into the YOLO series greatly enhances the performance.

Given the disparities between regular objects and those in RSIs, many researchers have focused on multiscale object detection to tackle the challenges presented by small objects in RSIs. Zhang et al. [16] designed a multistage enhancement network to achieve feature optimization for tiny objects from different stages. Wang et al. [17] proposed a multiscale feature augmentation network to improve the extraction of multiscale features from the object region. The feature-reflowing pyramid structure proposed by Wang et al. [18] aimed to boost the detection performance across various scales by incorporating detailed information from the adjacent lower levels. A novel informative feature pyramid network [19] has been introduced to tackle the problem of inadequate feature extraction. Peng et al. [20] presented a lightweight YOLO model that integrates attention mechanisms and multiscale features to tackle the difficulties posed by small objects in images. Zhou et al. [21] presented an improved spatial pyramid pooling model that effectively incorporates multiscale features, leading to a better capacity for the model to recognize objects.

Currently, many attention strategies have been implemented in RSOD to enhance the detection effectiveness. Lu et al. [22] introduced a dual-path attention module designed to enhance feature selection by employing both spatial and channel attention, which helps minimize background noise and highlight important features. Hu et al. [23] built a local attention module that incorporates deformable convolution to improve the extraction of local information for ships (SPs), thereby enhancing the network's robustness more effectively. An enhanced effective channel attention is proposed by Liu et al. [24] to refine feature representation and reduce the interference of complicated backgrounds.

Although CNN-based approaches have attained remarkable achievements in the field of RSOD, they are limited in their capacity to capture long-range dependencies. The Transformer architecture was initially designed for natural language processing [43]. Its core lies in the self-attention mechanism, which processes all positions in the input sequence in parallel, enabling the transformer to effectively handle long-range dependencies and adapt to input sequences of varying lengths. To overcome some of the constraints of CNNs, researchers have begun to explore the application of the method to computer vision tasks. Vision transformer [25] is a method used in image classification where the input image is divided into patches of a certain size. These patches are then spread and mapped into a vector space of a fixed dimension. Positional embeddings are added to these patches, and the resulting sequence of vectors is presented to the encoder for further processing. In the encoding stage, the self-attention mechanism establishes the relationship between the patches so that comprehensive contextual information can be captured. Introducing a novel perspective to image classification, vision transformer has sparked research into the processing of global and long-range semantic information. Consequently, other researchers have begun to incorporate the transformer into RSOD. For instance, a novel object detection network, SAFF-SSD, is proposed by Huo et al. [26] that designs a novel self-attention module named the local lighted transformer (2L-Transformer) block and couples it with the EfficientNetV2-S backbone network to enhance the feature extraction ability. Wang et al. [27] designed a hybrid detector called MashFormer for multiscale perception. This approach complements the CNN-based feature extraction backbone with a Transformer, enabling more effective capture of long-range feature relationships and enhancing representational capacity in complex background scenarios. Chen et al. [28] proposed the MDCT detection model, which addresses the issue of losing object information in intricate backgrounds by including the Transformer at the neck of the network. Min et al. [29] presented the innovative architecture YOLO-DCTI, which employs the contextual transformer infrastructure for small object detection. However, self-attention mechanisms suffer from a critical limitation due to their computational complexity, which is $O(H^2W^2(C_{qk}+C_v))$, resulting in less efficient inference.

Beyond conventional CNNs and transformer-based approaches, spiking neural networks (SNNs) demonstrate significant potential in RSI analysis due to their biologically-inspired event-driven characteristics and demonstrated energy efficiency advantages. Specifically, Niu et al. [30] proposed

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an event-driven SNN incorporating threshold-following reset (TF-reset) and membrane potential modulation mechanisms for RSI classification, achieving lossless ANN-to-SNN conversion with superior energy efficiency and noise robustness. Duan et al. [31] developed a novel online adaptation framework for SNNs that enables efficient deployment on resource-constrained edge devices while maintaining adaptability to environmental variations. Notably, Yang et al. [32] proposed SeLHIB, the first self-supervised information bottleneck learning framework for SNNs, which reduces the average endpoint error by 33.78% on the MVSEC dataset while achieving 90.44% lower energy consumption compared to conventional ANNs. Despite their potential, SNNs encounter persistent challenges in RSOD applications. Primarily, the temporal dependency of spiking signals induces significantly higher latency than feedforward neural networks, compromising real-time detection capabilities. Moreover, the nondifferentiable nature of discrete spiking activation functions substantially reduces end-to-end training efficiency compared to standard backpropagation algorithms. Furthermore, current neuromorphic hardware platforms demonstrate inadequate energy efficiency ratios and computational density for large-scale SNN deployment. In light of these unresolved technical limitations, our study focuses on innovating within the established CNN-Transformer framework to achieve an optimal balance between detection accuracy and real-time performance. A remedy to the aforementioned issue is proposed in our study by the utilization of the squeeze-enhanced axial attention module. The methodology emphasizes the compression of the input feature map in both horizontal and vertical dimensions to generate reduced columns and rows. Furthermore, the self-attention is calculated using this condensed representation. This approach can decrease the computational cost to $O((H^2 + W^2)(C_{qk} + C_v))$, while simultaneously capturing local features by the depthwise separable convolution. This process allows for the efficient extraction of richer contextual information, leading to lower computational cost and improved inference speed without compromising performance. Furthermore, we design a unidirectional multiscale coupling module (UMCM) to enhance the model's capability to identify small objects. To address the issue of significant differences in object sizes in RSIs, we suggest a multirouting scale-adaptive fusion (SAF) method that aims to establish an effective correlation between features and achieve a balanced detection for objects at various scales.

The main contributions of our research are summarized as follows.

1) Based on axis-squeeze and detail enhancement methods, the squeeze-enhanced axial attention module focusing on background perturbations is constructed, which can effectively aggregate global and local information, mitigate the interference of complicated background noise on the object, and reduce the computational burden in the process.
2) A novel approach is proposed for coupling fine-grained features with advanced features using a multirouting SAF strategy with multiple feed-forward connections, which enhances the synergy between features of varying sizes and allows for a compromise between detecting small and large objects.
3) We suggest an axis-squeeze and multirouting SAF network for RSOD. In this network, an UMCM has been developed for optimizing the object feature extraction capability and improving the small object detection accuracy. Results from ablation and comparison experiments on RSOD, VisDrone, and Dior datasets demonstrate the effectiveness of the proposed module and network.

The subsequent sections of this work are organized as follows. Section II includes an overview of the related work. Section III presents a comprehensive explanation of the suggested methodology. Section IV provides an introduction to the datasets and evaluation metrics employed in the experiments, and offers a detailed analysis of the results. The conclusion can be found in Section V.

II. Related Work

A. Multiscale Feature Extraction

Conventional CNNs rely on single-scale feature extraction and use multibranch standard convolutions to obtain multiscale features. However, there are two concerns with this strategy. On the one hand, depending solely on single-scale feature extraction may result in insufficient extraction of both global and local information due to the differences in object sizes. On the other hand, traditional standard convolutions have certain limitations when dealing with many elongated targets present in RSIs. By employing convolution operations of various sizes on multiple branches, multiscale feature extraction may acquire comprehensive features of objects without altering the input feature map scale. For example, Liu et al. [33] significantly enhance the ability to capture information across various object scales by integrating feature extraction modules at each level. This is achieved by through the use of atrous convolutions with varying dilation rates, which significantly enhance multiscale features. A multireceptive-field feature extraction module with atrous convolution is proposed in [34]. The module aims to aggregate the contextual information from several receptive fields, thereby improving the feature representation and enabling effective adaptation to scale variations of objects. To address the problem of conflicting information at various scales, BCEMS [35] is introduced, which allows the model to improve the integration of contextual information gathering by using multiple atrous convolutions with modified dilation rates. A multiscale feature extraction module [37] has been developed to extract rich and valuable features by performing convolution operations on multiple branches of varying scales. To improve the multiscale perception capability of the network, Zhang et al. [38] presented a feature extraction module that incorporates 5×5 and 1×1 convolution branches derived from the residual module.

However, the aforementioned methods primarily concentrate on extracting and improving semantic information via atrous convolution and multibranch standard convolution. During image border processing, atrous convolution may omit pixels near the edge, leading to the loss of boundary information. RSIs often includes numerous elongated objects, and the

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application of standard convolution has specific inherent constraints. Therefore, to address these constraints, we design an UMCM as a component of our proposed semantic information enhancement architecture to optimize the capability of feature extraction and improve the small objects detection accuracy.

B. Multiscale Information Fusion

CNN-based object detection models present differential details characteristics across features at various layers. Shallow features usually contain more fine-grained information, which makes them well-suited for detecting dense and small-sized objects. Whereas deep features are enriched with increased semantic features making them more appropriate for recognizing significant and larger objects. Feature pyramid networks [39] innovatively adopts a top-down path to convey rich semantic information from deeper feature maps to shallower ones, enhancing the expression of features at multiple scales and significantly improving detection accuracy. After that, many variants based on FPN have gradually emerged. Notable examples include the path aggregation network [40], bidirectional FPN [41], and feature enhanced pyramid network [42] construct various different information paths to efficiently aggregate multiscale features. In recent years, several improved FPN algorithms for RSOD have achieved remarkable results. By way of illustration, Xie et al. [44] proposed the latent feature pyramid network to tackle the deficiency of long-range dependency in feature fusion. Min et al. [45] presented a novel attention feature pyramid network, which is utilized to address the challenge of detecting low-resolution and densely distributed small objects in complex scenes. Chen et al. [46] designed the scene contextual feature pyramid network (SCFPN), which aims to strengthen the relationship between objects and their surrounding scenes while tackling the challenges associated with variations in object sizes. To integrate fine-grained details from shallow feature maps with semantic information from deeper feature maps, a novel bidirectional dense feature pyramid network [37] is proposed, achieving efficient multiscale feature fusion by expanding the size of the feature pyramid network and introducing skip connections.

Our work presents a multirouting SAF method that aims to further couple fine-grained features with high-level features using multiple feed-forward connections, enhancing the synergy between features of varying sizes and balancing the detection effectiveness for small and large-scale objects.

C. Small Object Detection in RSIs

Small object detection has become increasingly challenging in the field of computer vision. Chen et al. [47] defined small objects as those where the median relative area (i.e., the ratio of the bounding box area to the image area) across all instances of the same category falls within the range of 0.08%–0.58%. First, small objects occupy fewer pixels in an image, which makes them less distinctive in the image and difficult to be detected by algorithms. Second, as the network deepens, the features associated with small objects can be diminished in the deeper layers, resulting in degradation of detection performance. Furthermore, intricate backgrounds might pose challenges in detecting tiny objects, as the presence of noise interference in the background might detrimentally impact the accuracy of the detection method. To address these challenges, researchers adopt various strategies. Zhang et al. [48] proposed the feature enhancement module, feature fusion module, and spatial context awareness module. These modules enhance the network's ability to recognize local contexts, fuse features across multiple scales, and global correlation across channels and spaces, leading to enhanced feature representation of small objects. Zhang et al. [49] designed the feature enhancement module, which includes the feature aggregation structure and the attention generating structure. This module reduces background interference by leveraging multiscale contextual information and incorporating a coordinate attention mechanism to improve detection accuracy. Zhang et al. [50] proposed a hybrid attention module based on the Swin transformer, which enhances object detection in the transformer by establishing pixel-level correlations. This enables the network to precisely recognize small objects in intricate backgrounds. Lu et al. [51] used a cross-shaped window transformer as the backbone network to extract image features at multiple levels. The global feature fusion module [52] is a module that enhances the recognition of small objects through a global attention mechanism. It achieves this by enhancing the network's selectivity of spatial information during the feature fusion process. Li et al. [53] introduced a novel global and local attention mechanism that enhances object detection capabilities by combining local features with global contextual information.

Although CNNs excel at extracting local features, they encounter challenges in acquiring information regarding long-range dependencies. Transformer exhibit robust global contextual modeling capabilities, but they often present significant memory and computational requirements, especially when dealing with high-resolution RSIs. To address these problems, we employ the squeeze-enhanced axial attention module, which allows the network to efficiently capture and analyze detailed contextual information while minimizing computational resource usage.

III. Methodology

The section provides an overview of the structure of our proposed network, named axis-squeeze and multirouting SAF network (AMSFNet). The improved backbone of the model is depicted in Fig. 2, while the overall architecture is illustrated in Fig. 3. A detailed description of the three primary components can be found below: the UMCM, the multirouting SAF method, and the squeeze-enhanced axial attention module.

A. Unidirectional Multiscale Coupling Module

In order to better extract target features, especially in complex RSI environments containing a large number of elongated objects, we propose the use of UMCM as the feature extraction module and enhance feature representation through a channel attention mechanism, as illustrated in Fig. 2. The UMCM adopts a multibranch structure, where each branch focuses on capturing features at a certain scale, leading to a rich and discriminative set of semantic features. Specifically, the

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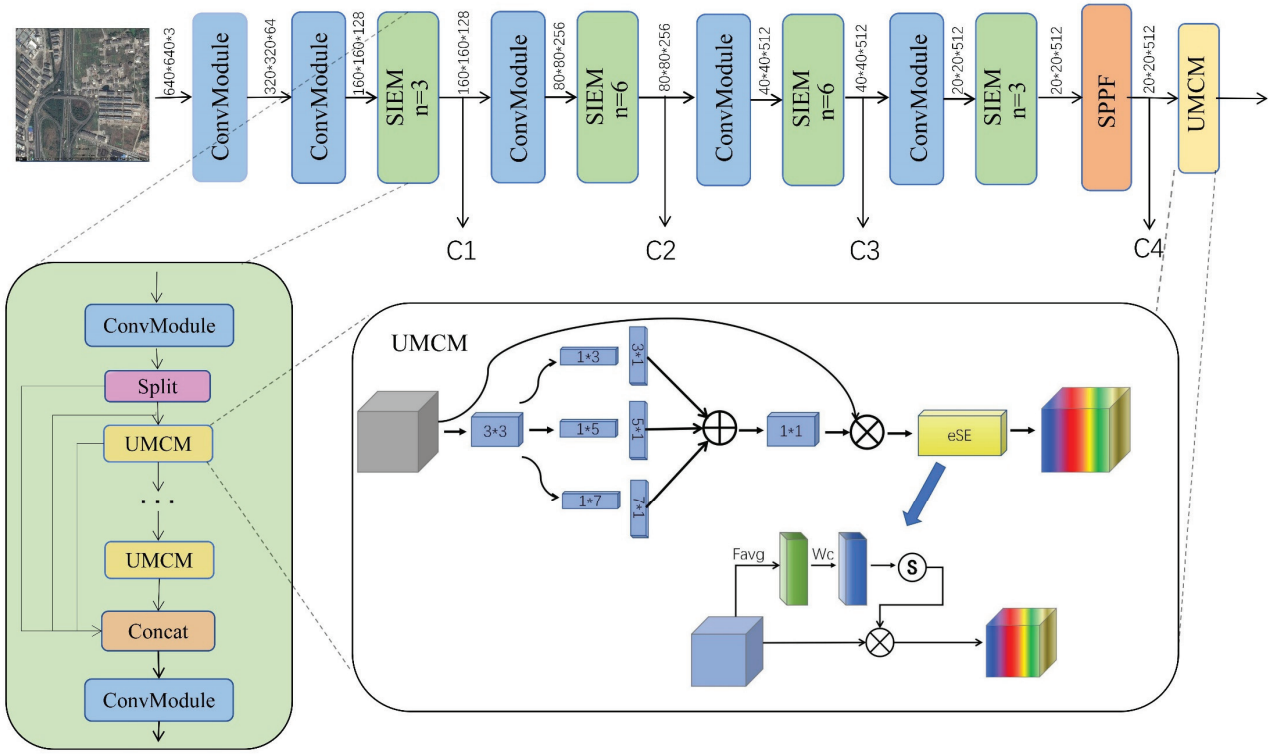


Fig. 2. Structure of our backbone, which has four stages, each with a convolution and a semantic information enhancement module. The details of the UMCM have been illustrated.

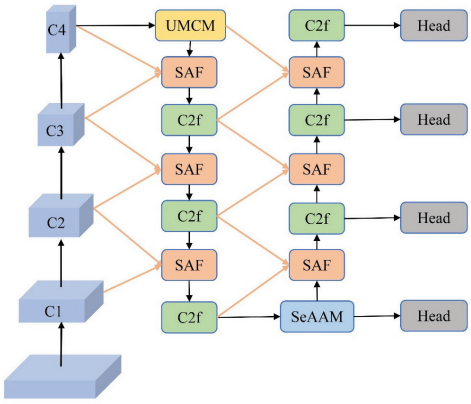


Fig. 3. Overall architecture of our model, where c1, c2, c3, and c4 come from the outputs of different stages, respectively.

early feature map undergoes a 3 × 3 convolution, followed by unidirectional convolution at various scales to extract features through three branches. Each output of the branches is then concatenated along the channels, and restore the number of channels using 1 × 1 convolution.

However, the use of unidirectional convolution may be inadequate when dealing with nonhorizontal and nonvertical features. Hence, we adjust the weight of the contribution of different channels by introducing a lightweight and efficient eSE module [54]. The eSE module first compresses the spatial information of each channel into a global feature through global average pooling (GAP), effectively capturing the contextual information of the feature map. This is particularly suitable for handling irregularly shaped or elongated targets, as these targets may exhibit significant variations in different directions. Then, a fully connected layer is used to learn the dependencies between channels, generating channel weights that dynamically identify and enhance the feature channelsmore critical to the task. This ensures that the model maintains high detection accuracy even when the orientation of the target changes. Finally, the learned channel weights are applied to the original feature map, strengthening the representation of key channels. This enables the model to more precisely focus on the features most crucial for detecting the target, thereby improving overall performance. Consequently, the issue of directional bias can be alleviated, allowing the network to demonstrate stronger adaptability when processing data from various orientations. The mathematical representation of these operations can be demonstrated as

$$F' = f_{\text{Conv}}^{3\times3}(F) \tag{1}$$

$$W_1 = f_{\text{Conv}}^{3\times1}\left(f_{\text{Conv}}^{1\times3}\left(F'\right)\right) \tag{2}$$

$$W_2 = f_{\text{Conv}}^{5\times1}\left(f_{\text{Conv}}^{1\times5}\left(F'\right)\right) \tag{3}$$

$$W_3 = f_{\text{Conv}}^{7\times1}\left(f_{\text{Conv}}^{1\times7}\left(F'\right)\right) \tag{4}$$

$$F'' = \left[f_{\text{Conv}}^{1\times1}\left[\text{Cat}\left(W_1, W_2, W_3\right)\right]\right] \odot F \tag{5}$$

$$Y = S\left(Wc\left(\text{Favg}\left(F''\right)\right)\right) \odot F' \tag{6}$$

where $f_{\text{Conv}}^{3\times3}, f_{\text{Conv}}^{3\times1}, f_{\text{Conv}}^{1\times3}, f_{\text{Conv}}^{5\times1}, f_{\text{Conv}}^{1\times5}, f_{\text{Conv}}^{7\times1}, f_{\text{Conv}}^{1\times7}$, and $f_{\text{Conv}}^{1\times1}$ represent 3 × 3, 3 × 1, 1 × 3, 5 × 1, 1 × 5, 7 × 1, 1 ×, and 1 × 1 convolution operation, respectively. Cat is the feature map concatenation operation, while F denotes the input feature map. ⊙ denotes the element-wise multiplication.

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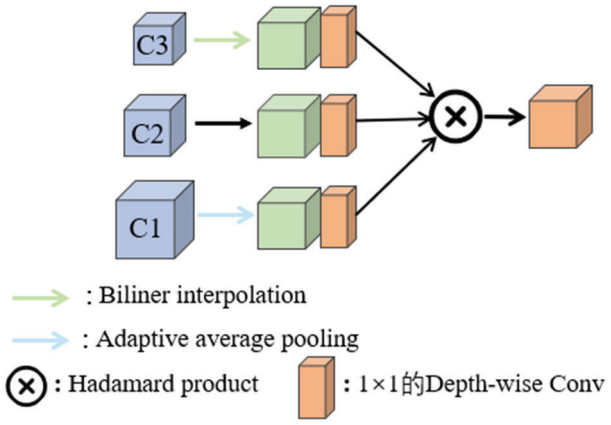


Fig. 4. Structure of the SAF.

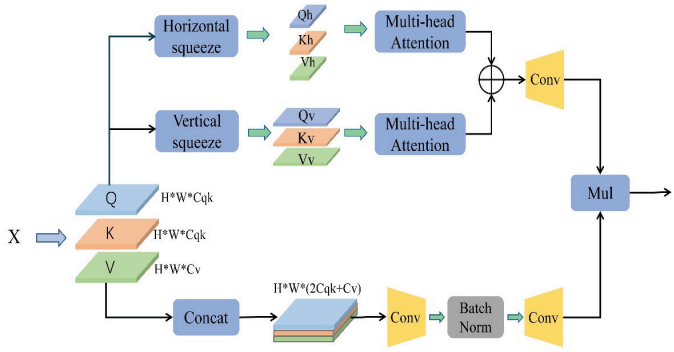


Fig. 5. Squeeze-enhanced axial attention module.

W_1, W_2, and W_3 denote the output feature maps after the three branches. S is the Sigmoid activation function and Wc represents the fully connected layer, respectively.

We integrate the standard ConvModules and our improved SIEMs consisting of UMCMs at each stage, as shown in Fig. 2. Differences exist in the number of SIEMs employed at different stages. The outputs generated by the first three SIEMs are used as inputs to the feature fusion in the following neck structures, C1, C2, and C3, respectively. Besides, the output of the fourth SIEM is passed on to the spatial pyramid pooling-fast (SPPF) for building C4.

B. Multirouting SAF Method

To effectively integrate shallow-layer fine-grained features with deep-layer semantic representations, we introduce an enhanced multirouting SAF module (MSFM), as illustrated in Fig. 3.

In MSFM, we make improvements in two main ways: the connection strategy is modified, and our approach fuses features from multiple stages, therefore facilitating the capture of objects at various scales. Furthermore, we modify the fusion strategy and adopt the structure of SAF, as depicted in Fig. 4. The above enhancements facilitate the collaboration of features of varying sizes and make efficient connections across features, therefore allowing the network to capture crucial information from the entire image.

The SAF component is shown in Fig. 4. Using C1, C2, and C3 obtained from the backbone as examples, we treat C2 as the object feature map size. For C1, the size of the current feature map is larger than the object size. Adaptive average pooling is employed to downsample the feature map to the object size, and the number of parameters can be reduced while enhancing the ability to extract features. For C3, the present feature map is smaller than the object size. To enlarge it to the object size, bilinear interpolation can be used. This method effectively preserves the detailed information in the image. Given below are the mathematical descriptions

$$f_{ij} = \begin{cases} \mathrm{S}(f_j, (H_i, W_i)), & \text{if } j < i \quad (7) \\ I(f_j), & \text{if } j = i \quad (8) \\ A(f_j, (H_i, W_i)), & \text{if } j > i \quad (9) \end{cases}$$

where S, I, and A represent bilinear interpolation, identity mapping, and adaptive average pooling, respectively. H_i and W_i represent the dimensions of the object feature map f_i, indicating its height and width. Besides, f_j denotes the feature map at level j. Upon using a 1×1 convolution for resampling

$$f'_{ij} = f_{\mathrm{Conv}}^{1\times 1}\left(f_{ij}\right). \quad (10)$$

Subsequently, the resampled feature maps undergo elementwise Hadamard product to further enhance each feature map, giving rise to additional semantic information and finer detailing, as demonstrated below

$$f''_i = H\left(\left[f'_{i1}, f'_{i2}, f'_{i3}\right]\right) \quad (11)$$

where Hdenotes the Hadamard product and then f''_i is passed through C2f before being used in the next stage of adaptive fusion. By fusing features at several levels, a more comprehensive feature representation can be constructed, which contributes to improving object detection performance in RSIs with complicated backgrounds and multiscale objects.

C. Squeeze-Enhanced Axial Attention Module

Accurate recognition of small objects in RSIs might be hindered by the presence of an intricate background, as spectral and textural noises in the background can interfere with the effectiveness of the detection algorithms. In the present research, we introduce axial squeeze attention to extract global semantic information and enhance local details using efficient convolution-based approaches. Combining CNN with Transformer allows for the simultaneous utilization of both local details and global context, hence improving the model's effectiveness in detecting small objects.

As shown in Fig. 5, we first extract the query, key, and value from x using $W_q, W_k \in R^{C_{qk}\times C}$ and $W_v \in R^{C_v\times C}$. Then, we perform horizontal and vertical squeezes to get $Q_{(h)}, K_{(h)}, V_{(h)} \in R^{H\times C_{qk}}$, and $Q_{(v)}, K_{(v)}, V_{(v)}, \in R^{W\times C_{qk}}$. This operation reserves the global information to a single axis, thus greatly reducing the complexity of following global semantic extraction and minimizing redundant information from large background regions. Subsequently, multihead self-attention is performed in both the horizontal and vertical dimensions followed by element-wise addition. This approach decreases

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the computational cost to $O((H^2+W^2)(C_{qk}+C_v))$. The axial squeeze attention is represented in the following equation:

$$\mathbf{y}_{(i,j)} = \sum_{p=1}^{H} \text{softmax}_p\left(\mathbf{Q}_{(h)i}^{\top}\mathbf{K}_{(h)p}\right)\mathbf{V}_{(h)p} + \sum_{p=1}^{W} \text{softmax}_p\left(\mathbf{Q}_{(v)j}^{\top}\mathbf{K}_{(v)p}\right)\mathbf{V}_{(v)p}. \tag{12}$$

The standard axial squeeze attention mechanism fails to account for positional awareness and disregards positional information present in the feature map. To enhance its effectiveness, we introduce position embeddings to enrich the axial squeeze attention. More precisely, the position embeddings $r_{(h)}^{q}$, $r_{(h)}^{k} \in R^{H\times C_{qk}}$ are coupled to make $Q_{(h)}, K_{(h)}$ aware of their positions within the axial squeeze features. Position embeddings are derived by linear interpolation from learnable parameters $B_{(h)}^{q}, B_{(h)}^{k} \in R^{L\times C_{qk}}$, where L is a constant. Similarly, $r_{(v)}^{q}$, $r_{(v)}^{k} \in R^{W\times C_{qk}}$ are also applied to $q_{(v)}, k_{(v)}$. For small objects, spatial position information is critical for detection. Position embeddings explicitly encode spatial coordinates into feature representations, allowing the model to accurately localize small objects within the image. Even when object features demonstrate high similarity to background noise, these embedded positional cues provide discriminative spatial information that effectively distinguishes objects from background interference, consequently enhancing detection accuracy. The position-aware axial squeeze attention is explicitly represented in the following equation:

$$\mathbf{y}_{(i,j)} = \sum_{p=1}^{H} \text{softmax}_p\left(\left(\mathbf{Q}_{(h)i}+\mathbf{r}_{(h)i}^{q}\right)^{\top}\left(\mathbf{K}_{(h)p}+\mathbf{r}_{(h)p}^{k}\right)\right)\mathbf{V}_{(h)p} + \sum_{p=1}^{W} \text{softmax}_p\left(\left(\mathbf{Q}_{(v)j}+\mathbf{r}_{(v)j}^{q}\right)^{\top}\left(\mathbf{K}_{(v)p}+\mathbf{r}_{(v)p}^{k}\right)\right)\mathbf{V}_{(v)p}. \tag{13}$$

While the squeezing operation efficiently extracts global semantic information, it neglects local details. To increase the spatial details, an auxiliary convolution-based kernel is used, as depicted in Fig. 5. The query, key, and value generated from X are concatenated along the channel dimension and then processed through 3×3 convolutions, followed by batch normalization. By using 3×3 convolutions, supplementary local information can be integrated from the query, key, and value. The 3×3 convolution can capture these fine-grained features by aggregating information from adjacent pixels, enhancing the distinguishability of small objects against background interference. Following that, 1×1 convolution is employed to decrease the dimensions from $(2C_{qk}+C_v)$ to C and produce weights for detail enhancement. Ultimately, the weights for enhancing details are multiplied with the features generated by the axial squeeze attention. The proposed squeeze-enhanced axial attention module can be represented as follows:

$$Y = \left[f_{\text{Conv}}^{1\times 1}\left(f_{\text{Conv}}^{3\times 3}\left(\text{Cat}\left(q,k,v\right)\right)\right)\right]\odot \mathbf{y}_{(i,j)} \tag{14}$$

where $f_{\text{Conv}}^{1\times 1}, f_{\text{Conv}}^{3\times 3}$ represent 1×1 and 3×3 convolutions, respectively. Cat denotes concatenation along the channel dimension, $\mathbf{y}_{(i,j)}$ represents the axial squeeze attention. The comprehensive integration of global semantic extraction and local detail enhancement is achieved by axial squeeze and detail expansion resulting in reduced computational complexity and sustained superior performance.

IV. Experimental Results and Analyses

A. Datasets

1) DIOR Dataset [3]: This is the largest benchmark dataset currently used for object detection in optical RSIs. There are 20 distinct ground object categories, which include: airplane (AL), airport (AT), baseball field (BF), basketball court (BC), bridge (BG), chimney (CM), dam (DM), expressway service area (EA), expressway toll station (ES), harbor (HB), golf course (GC), ground track field (GF), overpass (OP), SP, stadium (SD), storage tank (ST), tennis court (TC), train station (TS), vehicle (VH), and windmill (WM). We set the ratio of the training set, validation set, and test set to 6:2:2.

2) RSOD Dataset [71]: It consists of 2326 images and is divided into four categories: ALs, playgrounds, overpasses, and oil tanks. Their distribution in the dataset is as follows: 446 images contain 4993 ALs, 189 images include 191 playgrounds, 176 images contain 180 overpasses, and 165 images contain 1586 oil tanks. We divide the dataset into training and test sets according to an 8:2 ratio.

3) VisDrone-DET2019 [72]: It is a large-scale dataset comprising images captured by drones across various scenes, containing numerous small objects. The dataset consists of a total of 8599 images, divided into 6471 for training, 548 for validation, and 1580 for testing.

B. Evaluation Metrics

We utilize the average precision (AP) for individual classes and the mean AP (mAP) as our evaluation metrics. Generally, a higher mAP value indicates better detection performance of the model. The expressions for AP and mAP are as follows:

$$\text{AP} = \int_0^1 P(R)\,dR \tag{15}$$

$$\text{mAP} = \frac{1}{N_{\text{Cls}}}\sum_{i=1}^{N_{\text{Cls}}}\text{AP}_i \tag{16}$$

where P and R represent the precision and the recall, respectively, and N_{Cls} indicates the overall count of classes. The calculation formulas for P and R are as follows:

$$P = \frac{\text{TP}}{\text{TP}+\text{FP}} \tag{17}$$

$$R = \frac{\text{TP}}{\text{TP}+\text{FN}} \tag{18}$$

where TP, FN, and FP represent the counts of true positives, false negatives, and false positives, respectively.

C. Implementation Details

Our experimental setup is configured on the Windows 11 operating system, utilizing the PyTorch framework and deployed on a workstation equipped with NVIDIA RTX 4090 GPUs. For loss optimization, we employed stochastic gradient

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TABLE I
ABLATION EXPERIMENT ON RSOD

Method	aircraft	Oil tank	overpass	playground	mAP50	Parameter	GFLOPs
Baseline	95.5	98.8	74.1	97.7	91.5	11.12	28.40
+ UMCM	96.0	98.0	81.9	98.0	93.5	11.40	29.40
+ MSFM	95.9	98.1	80.7	97.9	93.2	10.21	35.40
+SeAAM	95.1	98.1	80.4	98.4	93.0	11.51	32.00
+ UMCM + MSFM	97.4	98.4	84.8	98.4	94.8	10.48	36.40
+UMCM+SeAAM	95.4	97.5	80.7	98.6	93.0	11.20	41.50
+MSFM+SeAAM	97.2	97.4	78.0	97.6	92.5	10.10	38.50
+ UMCM + MSFM + SeAAM	97.3	97.8	87.4	99.3	95.4	10.45	39.80

TABLE II
ABLATION EXPERIMENT ON DIOR

Method	AT	TS	TC	WM	BG	EA	HB	SD	BC	GF	SP	ES	VH	AL	BF	ST	DM	GC	CM	OP	mAP50
Baseline	90.3	75.4	97.1	96.0	60.8	98.2	73.9	96.8	94.4	84.1	95.0	88.7	68.9	97.3	96.9	89.8	78.3	92.4	90.7	73.8	86.9
+ UMCM	90.7	79.4	97.8	95.4	61.7	98.2	75.4	96.3	94.5	87.6	95.1	89.2	69.7	97.4	96.6	89.8	81.4	93.2	89.5	73.8	87.6
+ MSFM	90.2	76.7	97.6	96.3	60.7	97.4	76.0	96.0	94.4	86.0	96.2	89.0	71.9	97.9	96.6	92.6	80.9	92.7	88.9	72.4	87.5
+SeAAM	92.3	78.6	97.1	96.6	62.5	98.2	74.9	96.6	94.8	89.6	95.6	89.9	69.8	97.9	96.8	90.5	81.9	93.0	92.1	74.6	88.2
+ UMCM + MSFM	89.4	78.1	97.7	96.5	62.9	98.0	75.1	96.3	93.8	88.4	96.0	89.3	72.1	97.9	97.0	92.7	81.0	93.1	90.1	73.5	87.9
+UMCM+SeAAM	92.2	79.9	98.0	97.5	63.6	98.7	76.7	96.4	95.1	86.7	95.7	89.8	73.5	97.9	97.3	92.9	82.1	92.9	91.2	74.6	88.6
+MSFM+SeAAM	91.5	78.4	97.9	96.9	61.5	97.7	74.4	96.0	94.7	87.6	96.1	89.5	72.0	97.8	96.4	93.0	77.9	92.1	88.7	73.8	87.7
UMCM + MSFM+SeAAM	93.2	76.3	98.1	96.8	66.6	97.3	74.8	96.5	92.2	91.3	95.8	89.4	75.3	97.9	96.5	92.8	83.8	92.6	92.3	73.6	88.7

TABLE III
ABLATION EXPERIMENT ON VISDRONE

Method	pedestrian	people	bicycle	car	van	truck	tricycle	awning-tricycle	bus	motor	mAP50
Baseline	27.7	14.6	9.9	71.6	37.5	37.9	18.0	17.9	56.3	29.9	32.1
+ UMCM	28.1	15.0	10.0	72.4	38.2	40.0	18.3	19.9	56.5	30.0	32.8
+ MSFM	34.1	22.9	11.3	75.6	39.6	36.8	17.7	19.7	56.2	34.7	34.9
+SeAAM	27.9	15.1	10.1	72.0	37.1	39.6	18.0	19.2	56.9	29.9	32.6
+ UMCM + MSFM	34.7	22.0	11.8	75.7	39.7	37.5	19.8	21.4	57.1	34.2	35.4
+UMCM+SeAAM	34.9	23.9	12.5	75.7	40.0	39.3	20.9	20.6	59.1	34.8	36.2
+MSFM+SeAAM	33.7	22.3	12.7	76.4	41.2	39.7	19.0	19.5	58.5	34.0	35.7
+UMCM+ MSFM +SeAAM	34.8	23.4	13.1	76.5	41.4	38.9	20.2	23.1	58.8	35.7	36.6

descent (SGD), with an initial learning rate of 0.01, weight decay set to 0.0005, and momentum at 0.937. We use binary cross-entropy loss as the classification loss, while adopting distribution focal loss (DFL Loss) and complete intersection over union loss (CIOU Loss) for the bounding box regression loss.

D. Ablation Experiments

We conducted three ablation experiments on the RSOD, DIOR, and VisDrone datasets to validate the effectiveness of the proposed modules. Tables I–III display the results of the three ablation experiments.

1) Baseline + UMCM: This component enhances the feature representation capability of objects and improves the model's effectiveness in detecting small objects. In the VisDrone dataset, the AP of every category is improved. In RSOD, the AP for overpasses is increased by 7.8%, and in the DIOR dataset, the DM category makes a 3.1% improvement, and a 1.5% improvement for the HB category. This is strongly connected to the utilization of unidirectional convolutions at various scales that we implement.

2) Baseline + MSFM: The approach adopts a multirouting SAF strategy, promoting synergy between features of different sizes, and allowing the network to gather essential information from the entire image. On the three datasets, the mAP50 reaches 93.2% (↑1.7%), 87.5% (↑0.6%), and 34.9% (↑2.8%), respectively. In VisDrone, the AP for the pedestrian, people, car, and motor categories reaches 34.1% (↑6.4%), 22.9% (↑8.3%), 75.6% (↑4%), and 34.7% (↑4.8%), respectively. Combining UMCM and MSFM, our method achieves significant improvements of 3.3%, 1%, 3.3%, 1%, and 3.3% in mAP50 on the RSOD, DIOR, and VisDrone datasets, respectively.

3) Baseline + SeAAM: The module can mitigate the effects of intricate backgrounds and enhance the model's detection capabilities. In DIOR, objects like the background, such as ATs, BCs, GCs, and GFs, have shown improvements, with ATs improving by 2% and GF by 5.5%. In addition, we further conducted a visual analysis of the results of the three modules on three types of samples, with the specific results shown in Fig. 6. The three types of samples are: Sample A (containing objects with significant scale variations), Sample B (small objects with interference), and Sample C (occluded small objects). From the red boxes in Fig. 6, it can be seen that compared to the baseline method, our proposed approach demonstrates significant advantages. It not only effectively handles objects with large-scale variations but also maintains a low false detection rate for small objects. For example, in Sample A, our method achieves more accurate

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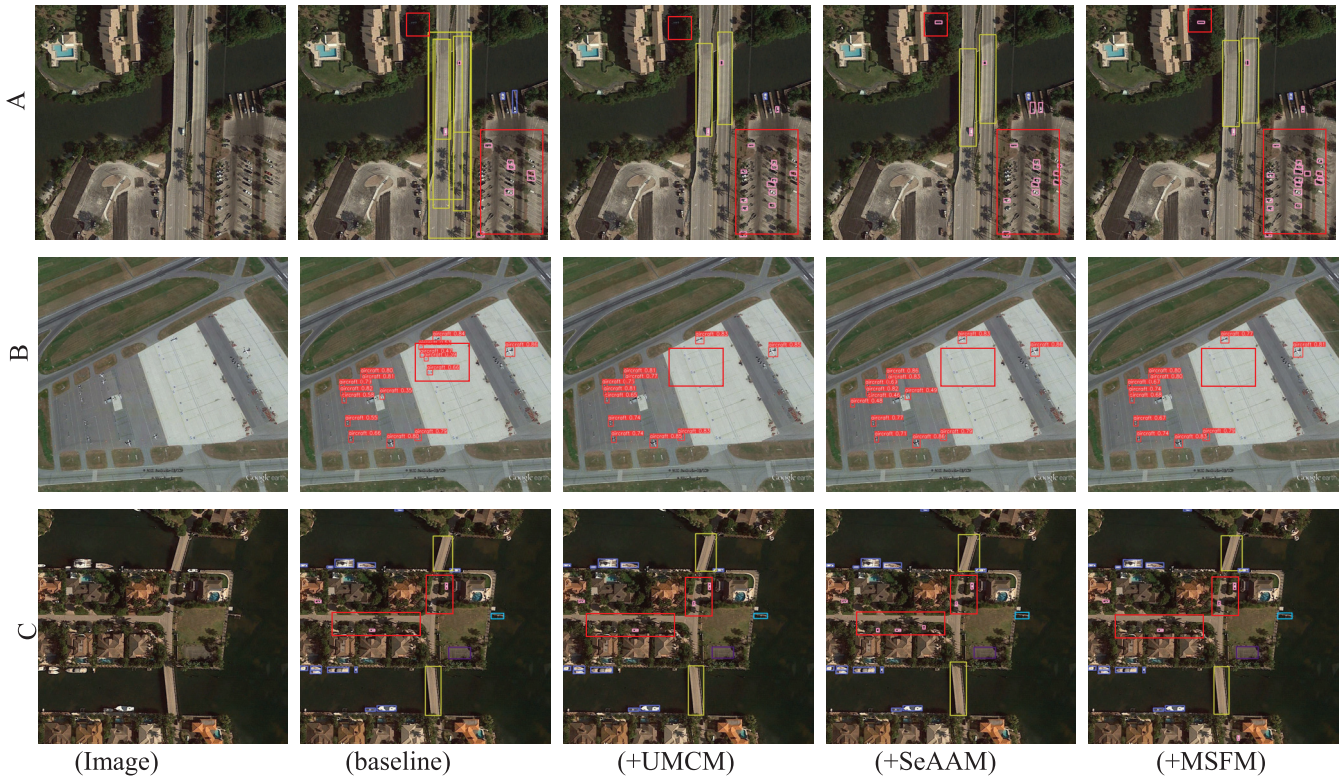


Fig. 6. Visualize the performance of individual modules in ablation studies across three different types of samples. Sample A contains objects with significant scale variations, such as bridges and cars; Sample B includes other objects that interfere with ALs; and Sample C consists of objects with occlusions.

recognition of objects at different scales; in Sample B, the baseline method produces numerous false detections, while our module significantly reduces false detection rates; in Sample C, even when objects are heavily occluded, our module can still successfully identify them, fully demonstrating its reliability in complex scenarios.

4) UMCM + MSFM: On the RSOD dataset, the mAP of aircraft reaches 97.4%, representing a 1.9% improvement over the baseline. On the DIOR dataset, it also performs impressively in scenarios with high background complexity or occlusions: the mAP of VH reaches 72.1% (↑3.2%), GFs reaches 88.4% (↑4.3%), and DM reaches 81% (↑2.7%). These results fully demonstrate the performance gains of this combination in complex environments.

5) UMCM + SeAAM: It performs prominently in challenging scenarios. The mAP of the train station(TS) reaches 79.9% (↑4.5%), and the mAP of the bridge (BG) reaches 63.6% (↑2.8%). For VHs with occlusions, their mAP reaches 73.5% (↑4.9%). On the VisDrone dataset, the precision of pedestrians and people, which are significantly affected by occlusion, reaches 34.9% and 23.9%, respectively, with increases of 7.2% and 9.3% compared to the baseline.

6) MSFM + SeAAM: On the DIOR dataset, for densely arranged and severely occluded SPs, the mAP reaches 96.1% (↑1.1%). On the VisDrone dataset, in scenes with dense occlusions, the mAP values of pedestrians and cars are 33.7% and 76.4%, respectively, with increases of 6% and 4.8% compared to the baseline, highlighting the optimized detection performance for highly occluded and dense objects.

Furthermore, when combining UMCM, MSFM, and SeAAM, the mAP on the RSOD, DIOR, and VisDrone datasets reaches 95.4%, 88.7%, and 36.6%. Compared to the baseline, this represents increases of 3.9%, 1.8%, and 4.5%, respectively. Meanwhile, the amount of parameters decreased by 0.67, but GFLOPs increased by 11.4. However, the AP for the oil tank category in the RSOD dataset is decreased.

This might be due to oil tanks being square-shaped objects, exhibiting similar features in both the horizontal and vertical dimensions. Such a constraint may hinder the UMCM and SeAAM from fully capitalizing on their benefits, leading to suboptimal optimization detection performance. Further enhancement could be undertaken.

E. Comparison of MSFM With Other Methods

To assess the effectiveness of the multirouting SAF method, we compare several regular FPN structures on the RSOD dataset. FPN, PANet, BiFPN, and AFPN may suffer from information loss or redundancy during feature fusion, leading to lower feature fusion efficiency, while the complex connection paths and weighting mechanisms also increase computational complexity. We enhance feature aggregation efficiency through multistage feature fusion and introduce a scale-adaptive mechanism to dynamically adjust feature map scales, preserving more detailed information while reducing computational complexity. To guarantee an equitable comparison, we set the number of detection heads to three. As shown in Table IV, MSFM achieves 82.7% in the OP category, surpassing the second-best result by 3.5%. Although MSFM does not achieve the best performance in all categories, it

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TABLE IV
COMPARISON OF MSFM WITH OTHER STRUCTURES ON RSOD

Methods	aircraft	oil tank	overpass	playground	mAP50	Parameter	GFLOPs	FPS
Baseline with FPN	94.6	98.3	79.2	97.8	92.5	20.37	76.5	56.54
Baseline with PANet	**95.5**	**98.8**	74.1	97.7	91.5	11.12	28.4	63.50
Baseline with BiFPN	95.0	98.3	68.0	**98.7**	90.0	11.20	28.9	63.77
Baseline with AFPN	95.4	97.6	76.3	97.6	91.7	11.64	33.3	49.99
Baseline with MSFM	94.9	97.7	**82.7**	97.3	**93.1**	**10.72**	**27.6**	**65.14**

TABLE V
COMPARISON WITH STATE-OF-THE-ART ON RSOD

Method	aircraft	Oil tank	overpass	playground	mAP50	Parameter	GFLOPs
Faster R-CNN[11]	81.34	96.78	85.45	95.47	89.76	41.17	206.71
FFCA-YOLO[48]	95.30	97.80	70.00	97.50	90.20	7.12	51.8
TPH-YOLOv5[55]	94.90	98.60	85.70	98.40	94.40	27.32	66.8
Drone-YOLO[56]	96.29	97.70	84.10	98.00	94.00	10.82	37.3
YOLOv5[57]	94.40	97.30	80.30	98.70	92.70	20.86	47.90
YOLOv7[58]	96.50	98.20	81.50	99.00	93.60	36.53	103.3
YOLOv8[59]	95.00	97.70	74.50	99.20	91.60	25.84	78.70
SSAFNet[60]	95.75	98.39	84.66	92.50	92.82	-	-
CF2PN[61]	95.52	**99.42**	83.82	95.68	93.61	91.60	-
AGMF-Net[62]	96.02	99.02	82.43	99.70	94.30	-	-
LAR-YOLOv8[36]	94.80	98.20	85.40	98.90	94.30	26.30	134.5
AeroDetectNet[63]	96.20	98.10	85.30	98.40	94.50	**4.96**	**10.30**
DA²FNet[64]	95.72	97.73	**90.56**	95.12	94.78	-	-
ABNet[24]	91.49	96.14	89.61	99.44	94.17	42.83	141.37
MPS-YOLO[70]	97.14	93.13	87.59	95.26	93.28	93.16	208.14
Mamba YOLO[73]	94.70	97.80	75.70	98.20	91.60	21.80	49.60
YOLOv8(Swim Transformer)[76]	93.00	96.30	72.80	98.20	90.10	51.31	455.00
PR-Deformable DETR[75]	90.30	96.7	93.30	**100.00**	95.10	46.07	151.60
RT-DETR[74]	95.90	97.90	77.90	97.60	92.30	31.99	103.40
ours	**97.30**	97.80	87.40	99.30	**95.40**	10.45	39.80

TABLE VI
COMPARISON WITH STATE-OF-THE-ART ON DIOR

Method	AT	TS	TC	WM	BG	EA	HB	SD	BC	GF	SP	ES	VH	AL	BF	ST	DM	GC	CM	OP	mAP50	Parameter	GFLOP
CATNet[65]	92.0	73.5	92.2	91.1	58.3	93.2	68.8	82.0	92.2	88.3	84.1	80.2	46.7	89.3	84.7	70.8	80.8	85.8	84.9	68.8	80.6	37.45	103.5
RepDarkNet[66]	89.5	56.7	95.5	93.6	62.2	94.5	62.9	92.3	92.5	87.1	95.7	85.8	71.4	97.7	94.9	89.5	68.2	91.4	91.4	72.9	84.3	-	-
YOLOX-CA[67]	92.5	76.8	93.9	93.8	60.5	94.8	68.5	87.5	92.2	88.1	77.2	87.7	64.5	93.6	89.8	78.6	81.9	88.3	84.6	70.8	83.3	13.84	20.37
Drone-YOLO[56]	90.1	76.8	98.0	96.8	60.4	97.8	75.8	96.8	94.8	86.0	95.4	89.3	70.6	97.8	96.7	91.9	79.1	93.5	91.0	73.7	87.6	10.82	37.3
YOLOX[15]	90.4	73.5	94.5	93.7	58.4	94.6	68.2	89.1	92.2	84.0	77.6	87.7	63.5	92.9	90.2	77.9	76.8	86.5	84.6	69.5	82.2	8.94	26.65
TPH-YOLOv5[55]	84.8	54.8	96.9	92.2	55.6	91.2	67.9	93.7	88.6	76.8	93.8	78.2	62.4	97.1	94.0	89.0	75.1	88.8	89.1	66.0	81.8	27.32	66.8
FL-YOLOv5[68]	91.4	75.0	94.2	93.5	58.3	93.8	66.4	85.4	92.2	88.7	91.3	87.2	61.2	91.2	89.7	81.2	76.7	86.6	83.8	68.7	83.1	**6.84**	**15.5**
FFCA-YOLO[48]	74.6	63.0	97.6	97.0	60.2	91.9	64.6	92.5	**95.3**	76.6	**96.2**	**90.8**	73.9	97.7	**97.4**	**92.9**	61.0	85.6	88.5	68.4	83.3	7.12	51.8
LAR-YOLOv8[36]	92.5	78.8	97.9	96.0	60.6	**98.5**	74.4	**97.1**	94.6	86.0	95.2	87.1	68.3	97.2	96.4	89.6	78.9	**93.0**	89.7	73.8	87.3	26.30	134.5
YOLOv5[57]	85.7	77.0	97.0	95.8	62.3	96.3	71.9	93.9	94.4	85.5	94.5	**90.8**	70.0	97.2	96.9	88.1	77.7	91.9	90.0	73.3	86.5	20.86	47.90
YOLOv8[59]	90.3	75.4	97.1	96.0	60.8	98.2	73.9	96.8	94.4	84.1	95.0	88.7	68.9	97.3	96.9	89.8	78.3	92.4	90.7	73.8	86.9	25.84	78.70
Mamba YOLO[73]	93.1	**80.7**	97.7	**97.5**	64.1	97.6	**76.0**	96.9	94.4	90.3	95.6	90.2	71.1	97.8	96.8	91.1	83.5	92.5	90.7	**74.6**	88.6	21.80	49.60
YOLOv8(Swim Transformer)[76]	89.6	67.4	97.5	96.8	61.9	97.0	75.2	96.9	95.2	85.5	95.3	86.2	70.0	97.6	96.3	90.6	74.5	91.6	91.5	**72.7**	86.5	51.31	455.00
RT-DETR[74]	91.3	77.6	97.0	96.9	65.8	97.2	70.9	94.7	94.8	84.0	93.6	91.9	70.5	97.8	97.0	86.7	82.9	92.4	92.1	72.1	87.4	32.00	103.5
ours	**93.2**	76.3	**98.1**	96.8	**66.6**	97.3	74.8	96.5	92.2	**91.3**	95.8	89.4	**75.3**	**97.9**	96.5	92.8	**83.8**	92.6	**92.3**	73.6	**88.7**	10.45	39.80

delivers the optimal results in terms of mAP50, Parameter, GFLOPs, and frames per second (FPS), leading to a compromise between accuracy and speed with comprehensive performance.

F. Comparing With Existing Works

The section presents a comparative analysis of our proposed AMSFNet and other state-of-the-art networks on RSOD, DIOR, and VisDrone datasets. The results can be seen in Tables V–VII.

1) Results on RSOD: Our approach is compared with 19 other networks carrying on RSOD, and the results are elaborated in Table V. AMSFNet obtains the highest mAP of 95.4%, outperforming all competitors. In addition, our approach achieves the AP of 97.3% on the AL category, demonstrating the method's effectiveness in detecting small

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TABLE VII
COMPARISON WITH STATE-OF-THE-ART ON VISDRONE

Methods	pedestrian	people	bicycle	car	van	truck	tricycle	aw-tricycle	bus	motor	mAP50	Parameter	GFLOPs
CenterNet[13]	22.60	20.60	14.60	59.70	24.00	21.30	20.10	17.40	37.90	23.70	26.20	25.00	33.62
SSD[9]	22.90	11.60	7.30	63.20	29.90	33.10	11.70	11.10	49.80	19.10	25.30	**5.00**	**4.62**
Drone-YOLO[56]	34.50	21.30	10.20	74.70	38.10	37.60	19.10	19.20	57.20	34.40	34.60	10.82	37.3
TPH-YOLOv5[55]	32.00	19.90	11.60	75.20	38.20	40.00	20.10	19.70	58.40	27.80	34.30	27.32	66.8
FFCA-YOLO[48]	34.30	19.80	11.20	76.40	41.30	38.60	20.00	19.00	57.20	30.00	34.80	7.12	51.8
LAR-YOLOv8[36]	27.90	16.20	11.10	71.80	39.30	40.10	16.80	19.30	58.20	30.30	33.10	26.30	134.5
YOLOX[15]	28.85	21.02	12.14	68.61	31.25	33.99	20.99	17.66	47.00	30.69	31.22	8.94	26.65
VAMYOLOX[69]	33.97	**23.62**	**15.56**	72.19	36.68	35.74	20.36	17.35	52.90	33.61	34.20	9.69	60.35
YOLOv5[57]	28.30	17.50	7.86	70.80	34.80	32.30	15.90	16.20	53.70	26.60	30.40	20.93	48.10
YOLOv7[58]	31.10	18.60	11.30	76.10	40.00	**45.80**	**24.30**	21.90	57.60	31.60	35.80	36.53	103.3
YOLOv8[59]	29.40	15.30	11.50	72.00	39.80	43.90	16.00	19.00	59.60	31.20	33.80	25.86	79.10
Mamba YOLO[73]	30.00	16.40	11.20	74.40	40.80	43.10	18.70	21.50	**60.00**	32.70	34.90	21.80	49.60
YOLOv8(Swim Transformer)[76]	25.10	12.20	10.20	70.00	34.20	30.70	19.80	17.90	51.90	27.30	29.90	51.31	455.00
RT-DETR[74]	20.90	10.90	8.60	66.00	31.60	45.30	19.70	18.10	55.40	24.70	30.10	32.00	103.50
Ours	**34.80**	23.40	13.10	**76.50**	**41.40**	38.90	20.20	**23.10**	58.80	**35.70**	**36.60**	10.45	39.80

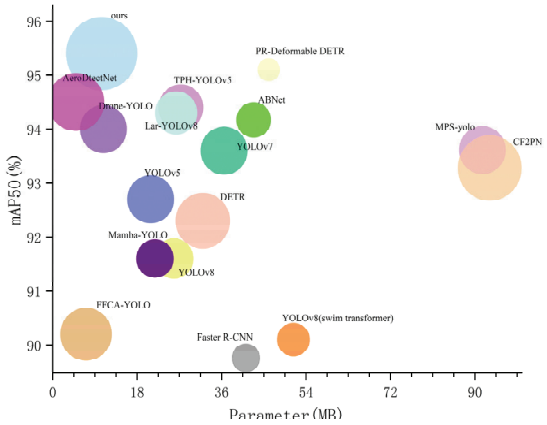


Fig. 7. Visual comparison of network performance evaluation.

objects. The number of parameters of our method is second only to FFCA-YOLO and AeroDetectNet. Furthermore, we perform an intuitive visual performance evaluation, as shown in Fig. 7. In the figure, the horizontal axis represents the model parameters, and the vertical axis is mAP50. The size of the circle in the figure indicates the AP of representative small-object ALs, and the larger the circle, the higher the accuracy.

2) Results on DIOR: The result of the AMSFNet is compared with 14 existing approaches using the current DIOR dataset and is provided in Table VI. AMSFNet outperforms all others with the greatest mAP of 88.7%. In eight of the 20 categories, it gets the best performance. Notably, AMSFNet performs exceptionally well on small objects such as ALs, SPs, and cars, with an AP of 97.9% for ALs, 95.8% for SPs, and 75.3% for VHs, which is 1.4% higher than the second-best result (73.9%), highlighting the effectiveness of our method for identifying small objects. Moreover, our model ranks fourth in terms of both parameters and GFLOPs, striking an excellent balance between performance and complexity.

3) Results on the VisDrone: We evaluate AMSFNet using the VisDrone dataset and present a comparative analysis with 14 other recent alternatives, as shown in Table VII. As desired, it receives the highest mAP of 36.6%, surpassing the aforementioned approaches. Furthermore, our approach attains the optimal outcomes in five specific categories: 34.80% AP for the pedestrian, 76.50% AP for the car, 41.40% AP for the van, 23.10% AP for the aw-tricycle, and 35.70% AP for the motor. The suggested method is highly efficient in identifying objects of different sizes. Especially in categories including a substantial quantity of small objects, such as pedestrians, cars, and motors, surpassing all other detectors in performance.

G. Comparison of Feature Visualization Results

We compare AMSFNet with YOLOv5, FFCA-YOLOv5, Tph-YOLOv5, Drone-YOLOv8, YOLOv8, LAR-YOLOv8, and Mamba-YOLO for feature visualization using the Grad-CAM method, and the results are shown in Fig. 8. AMSFNet exhibits better focusing capacity, which highlights the object's position and reduces the impact of background noise. In addition, for some very small objects, such as VHs near DMs, our method can also effectively focus on the objects without being interfered by other objects. Nevertheless, as was previously mentioned, AMSFNet does not highlight all oil tanks, thus, going forward, we will keep refining the related components in our future work.

H. Prediction Visualization Results

To further showcase the performance of AMSFNet, we present visualizations of its prediction results using several representative samples from the RSOD, VisDrone, and DIOR datasets. As shown in Fig. 9. AMSFNet realizes significant detection performance. In Fig. 8(a) and (b), the network can accurately recognize medium and large objects in the scene, demonstrating its remarkable adaptability to objects of various sizes. The improved detection of the challenging small objects in Fig. 8(c) gives complete validation to the advantages of our model in collecting details. Fig. 8(d) presents the prediction results in RSIs with complicated backgrounds. In Fig. 8(e), a range of objects with varying sizes are accurately identified. Despite the presence of several intricate backgrounds in the RSIs, our method remains very efficient in identifying objects inside them, particularly small-sized cars, SPs, and ALs. The presented visualization results indicate that our model once again exhibits exceptional performance in multiscale object detection on RSIs.

I. Inference Efficiency for Deployment Scenarios

Although the method proposed in this study has achieved the best results in terms of mAP50, which reflects accuracy,

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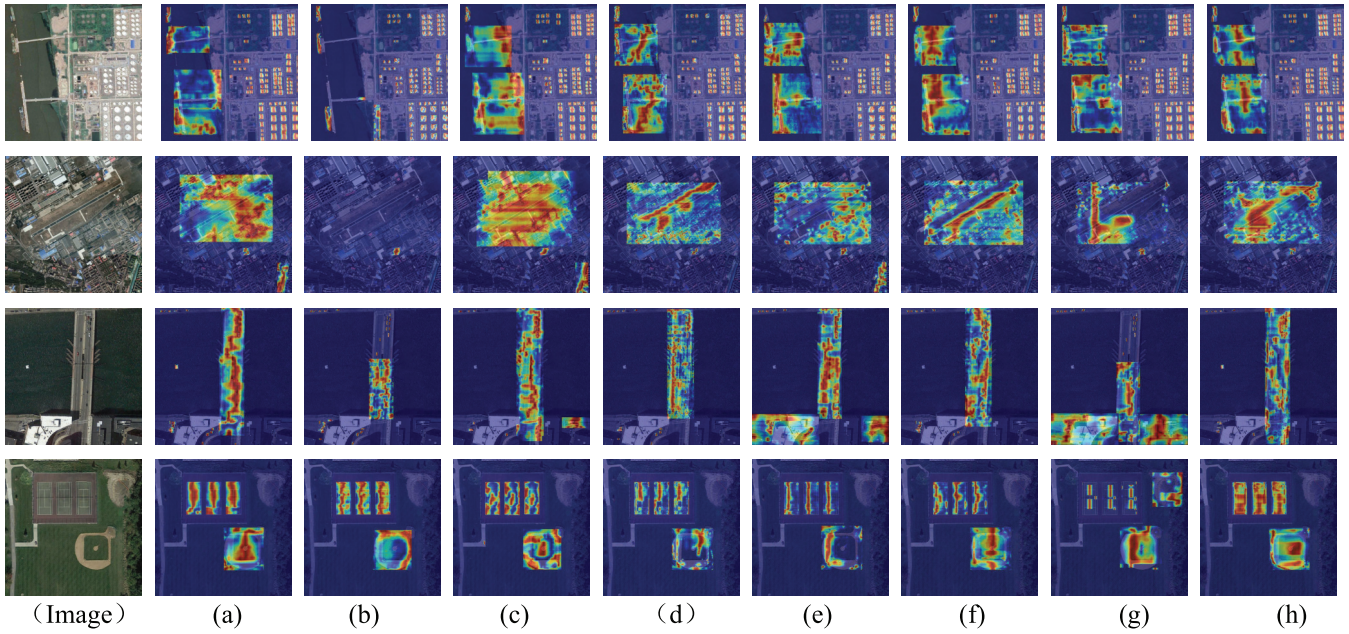


Fig. 8. Comparison of feature visualizations. (Image) Original image, (a)–(h) YOLOv5, FFCA-YOLOv5, Tph-YOLOv5, Drone-YOLOv8, YOLOv8, LAR-YOLOv8, Mamba YOLO, and AMSFNet, respectively.

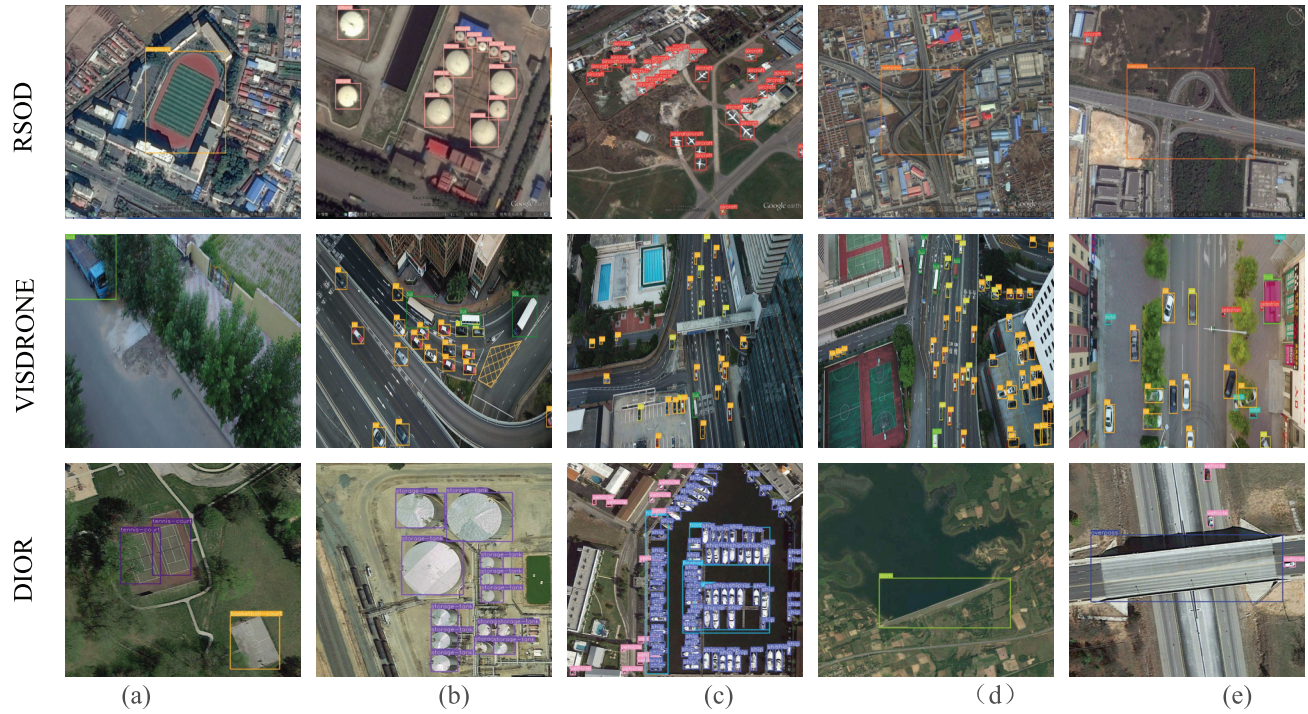


Fig. 9. Representative prediction results of AMSFNet on the RSOD, VISDRONE, and DIOR datasets, with considerable advantages in detecting objects across varying scales. For (a) large objects, (b) medium objects, (c) small objects, (d) objects in complicated backgrounds, and (e) multiscale objects within the image.

and has also demonstrated good performance in terms of the parameters and GFLOPs that reflect complexity, in order to further verify its real-time inference efficiency, we also compared the inference time, memory usage and FPS of our method on the conventional CPU that are more commonly used in edge devices, UAV systems and satellite platforms with several representative models, as shown in Table VIII. In this comparative analysis of inference time and memory usage and FPS, the AMSFNet model demonstrated reliable performance.

From the perspective of inference time, it exhibited stable and competitive performance across different input sizes. At smaller resolutions (640×640), although slightly slower than YOLOv5, YOLOv8, and FFCA-YOLOv5, it outperformed other models. As the resolution increased to 1280×1280 and 2560×2560, our model consistently showed faster inference

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TABLE VIII
COMPARISON OF INFERENCE TIME, MEMORY USAGE, AND FPS ON CPU FOR DIFFERENT MODELS

model	Inference Time (ms)			Memory Usage (M)			FPS		
	640*640	1280*1280	2560*2560	640*640	1280*1280	2560*2560	640*640	1280*1280	2560*2560
YOLOv8[59]	32.40	66.50	335.70	166.68	255.06	422.37	30.86	15.04	2.98
Lar-YOLOv8[36]	77.90	464.60	5306.90	209.73	362.57	648.72	12.84	2.15	0.19
Drone-YOLOv8[56]	70.70	174.00	1474.80	122.88	324.99	463.07	14.14	5.75	0.68
TPH-YOLOv5[55]	124.50	377.70	2192.40	102.55	332.12	476.51	8.03	2.65	0.46
FFCA-YOLOv5[48]	50.00	112.90	709.10	85.98	306.19	587.41	20.00	8.86	1.41
yolov8(Swim Transformer)[76]	90.70	320.30	1292.60	224.89	345.71	549.42	11.03	3.12	0.77
YOLOv5[57]	21.60	58.30	311.00	44.93	162.88	289.76	46.30	17.15	3.22
RT-DETR[74]	476.80	1683.20	7557.10	178.28	263.04	381.61	2.10	0.59	0.13
AMSFNet	70.70	236.30	1334.50	198.91	341.96	385.91	14.14	4.23	0.75

times, particularly excelling at larger resolutions where it significantly outperformed models like LAR-YOLOv8 and TPH-YOLOv5, highlighting its robust processing capability.

Regarding memory usage, when dealing with lower resolutions, the memory consumption of this model, including the slightly higher memory footprint for reading image data, was greater than that of some competing models. It is worth emphasizing that when the input size was enlarged to 2560×2560, the memory optimization ability of this model became remarkably prominent. In terms of memory usage, it ranked third, only surpassed by the outstanding YOLOv5 and RT-DETR.

Particularly when concentrating on the resolution scenario of 640×640, our proposed method managed to attain an inference speed of 14.14 FPS. Among the models that were compared, it ranked fifth. Although it is not the model with the highest inference speed, this performance is still adequate for applications that are close to real-time requirements for low-resolution images.

Although our method has not yet achieved optimal performance in inference efficiency, it demonstrates significant advantages in detection accuracy and complexity. In the future, we will further optimize the model to improve inference efficiency for large-scale real-time image deployment scenarios.

V. CONCLUSION

The present work introduces an innovative one-stage object detection network, AMSFNet, specifically developed to improve multiscale detection in RSIs, particularly for small-scale objects. By taking advantage of three crucial improvements, AMSFNet efficiently integrates global semantic information with fine-grained local feature details. The primary objective of the UMCM is to enhance the representation's extraction capabilities and ensure higher detection accuracy for small objects. Furthermore, the Squeeze-enhanced Axial Attention Module efficiently incorporates both global and local information to mitigate the interference caused by the intricate background on the object. Besides, the multirouting SAF approach can enhance the synergistic impact produced by features of varying sizes, thereby achieving a compromised detection for objects of both small and large scales. Experimental ablation studies provide evidence of the efficacy of each component of the suggested methodology.

The performance of AMSFNet is shown to be superior in comparison experiments conducted with consistent dataset settings across different detection methods. Despite the substantial enhancement in the detection performance of small objects, our approach may not be very effective in identifying some square objects, such as oil tanks, which requires further investigation. Furthermore, we will focus on optimizing the model's inference efficiency through computational latency reduction, while simultaneously exploring the integration of SNNs to enhance energy efficiency for edge deployment, thereby improving our model's adaptability to complex and dynamic real-world application scenarios.

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